

Example: 1

For the banana function $f(x, y) = (x - 1)^2 + 100(y - x^2)^2$, its gradient can be written as

$$\bar{\nabla} f = \begin{pmatrix} 2(x - 1) - 400x(y - x^2) \\ 200(y - x^2) \end{pmatrix} = 0,$$

$$2(x - 1) - 400x(y - x^2) = 0, \quad 200(y - x^2) = 0.$$

This means that $x_* = 1$ and $y_* = 1$.

$$f(x_*, y_*) = 0$$

Example: 2

Minimize: $-10 + x^2 + (y - 2)^2$

$$L(x, y, \lambda) = -10 + x^2 + (y - 2)^2 + \lambda(x + 2y - 5),$$

which becomes an unconstrained optimization problem for $L(x, y, \lambda)$.

$$\frac{\partial L}{\partial x} = 2x + \lambda = 0, \quad \frac{\partial L}{\partial y} = 2(y - 2) + 2\lambda = 0,$$

or

$$x = -\frac{\lambda}{2}, \quad y = -\lambda + 2.$$

$$x + 2y = 5, \quad \lambda = -2/5 \text{ is the right value}$$

$$\text{we got } x = 1/5 \text{ and } y = 12/5$$

Example 3

we can now define

$$L(x, y, \lambda_1, \lambda_2) = x + y^2 + \lambda_1(x^2 + 2y^2 - 1) + \lambda_2(x - y + 1).$$

The stationary conditions are

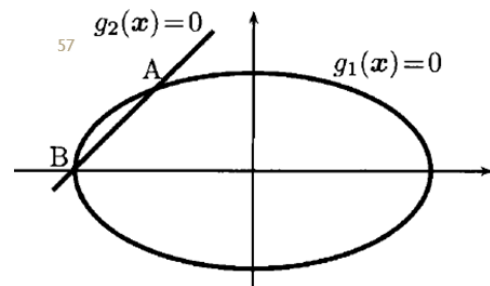
$$\frac{\partial L}{\partial x} = 1 + 2\lambda_1 x + \lambda_2 = 0, \quad \lambda_1 = -\frac{7}{6}, \quad \lambda_2 = -\frac{16}{9}, \quad \text{or} \quad \lambda_1 = \frac{1}{2}, \quad \lambda_2 = 0.$$

These correspond to the two points A(-1/3, 2/3) and B(-1, 0). The objective function takes the value 1/9 and -1 at A and B, respectively. So the best solution is $f_{\min} = -1$ at $x_* = -1$ and $y_* = 0$ (see point B in Figure 4.3).

$$\frac{\partial L}{\partial y} = 2y + 4y\lambda_1 - \lambda_2 = 0,$$

$$\frac{\partial L}{\partial \lambda_1} = x^2 + 2y^2 - 1 = 0,$$

$$\frac{\partial L}{\partial \lambda_2} = x - y + 1 = 0.$$



Example 4

First, we write the above problem in terms of the Lagrangian

$$L = (x - 2)^2 + 4(y - 3)^2 + v_1(-x + y - 2) + v_2(x + 2y - 3).$$

The KKT conditions become

$$2(x - 2) - v_1 + v_2 = 0, \quad v_1(-x + y - 2) = 0,$$

$$v_2(x + 2y - 3) = 0,$$

$$8(y - 3) + v_1 + 2v_2 = 0, \quad \text{and} \quad v_1 \geq 0, \quad v_2 \geq 0.$$

The first two conditions give

$$x = \frac{v_1 - v_2}{2} + 2, \quad y = -\frac{(v_1 + 2v_2)}{8} + 3.$$

The two complementarity equations imply four possibilities:

- If $v_1 = v_2 = 0$, we have $x = 2, \quad y = 3$,

which is essentially the solution without any constraint. However, $x = 2$, and $y = 3$ do not satisfy the constraint $x + 2y - 3 \leq 0$. So this solution is not feasible.

- If $v_1 = 0$ but $v_2 \neq 0$, we have $x + 2y - 3 = 0$.

$$\text{we have } v_2 = 5, \quad x = -1/2, \quad y = 7/4.$$

This set of solution does not satisfy $-x + y \leq 2$. So this is not a feasible solution.

- If $v_2 = 0$ but $v_1 \neq 0$, we have $x = v_1/2 + 2$ and $y = -v_1/8 + 3$.
 $-x + y = 2$.

This gives $v_1 = -8/5$ and subsequently leads to $x = 6/5$ and $y = 14/5$. However, this is not a feasible solution because it does not satisfy the other constraint $x + 2y \leq 3$.

- If $v_1 \neq 0$ and $v_2 \neq 0$, we have two equalities as the inequalities become tight $-x + y = 2, \quad x + 2y = 3$,

whose solution is $x = -1/3$ and $y = 5/3$.

$$\text{we have } v_1 - v_2 = -\frac{8}{3}, \quad v_1 + 2v_2 = \frac{32}{3}.$$

Its solution can be obtained easily as

$$v_1 = 16/9, \quad v_2 = 40/9,$$

which indeed satisfy $v_1, v_2 \geq 0$. This is also a feasible solution, and its objective function has a value of $f(-1/3, 5/3) = 113/9 \approx 12.556$.